

Ultimate Music Playback and Popularity Analysis,Atlantic (USA)

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Abstract

Music is one of the best therapies in the world. The expansion of digital music consumption has led to the availability of large-scale datasets that can be used to understand patterns in artist popularity and listener behavior. This study utilizes a dataset from Atlantic Music (United States), containing information on artists, albums, singles, historical popularity scores, and user listening time measured in hours. The objective is to analyze and predict the popularity of artists based on prior song performance and audience engagement using Python-based data analytics techniques.

The analysis is conducted using Python libraries such as Pandas and NumPy for data pre-processing and transformation, Matplotlib and Seaborn for exploratory data visualization, and Scikit-learn for implementing machine learning models. Exploratory Data Analysis (EDA) is performed to identify trends, correlations, and outliers within the dataset. Regression models, including Linear Regression and Random Forest Regressor, are applied to predict artist popularity, while clustering algorithms such as K-Means are used to group artists based on listening patterns and growth trajectories.

The results indicate that previous song popularity and total listening hours are strong predictors of current artist success. Machine learning models demonstrate reliable performance in forecasting popularity trends, highlighting the effectiveness of Python tools in music data analytics. This study emphasizes the importance of data-driven insights for strategic decision-making in the music industry and provides a foundation for future work in recommendation systems and real-time analytics.

1 Introduction

The digital transformation of the music industry has significantly changed how music is produced, distributed, and consumed. With the rise of streaming platforms, vast amounts of user-generated data are continuously being collected, providing valuable insights into listener behavior and artist performance. This data-driven environment has created new opportunities for applying data analytics and machine learning techniques to understand and predict music popularity.

Music-related data analytics focuses on extracting meaningful patterns from datasets that include artist information, album releases, single tracks, and user engagement metrics such as listening time. In particular, understanding how an artist's previous work influences their current popularity has become an important area of study for record labels, marketers, and streaming services. Factors such as historical hit performance, release frequency, and audience retention play a crucial role in shaping an artist's success.

This research utilizes a dataset from Atlantic Music (United States), which contains detailed information on multiple artists, their albums, individual songs, popularity scores, and total hours of listening. The aim of this study is to analyze the relationship between past performance indicators and current popularity levels, and to develop predictive models that can estimate an artist's future success.

To achieve these objectives, Python-based data analysis tools are employed. Libraries such as Pandas and NumPy are used for data preprocessing and manipulation, while Matplotlib and Seaborn

support data visualization and trend analysis. Machine learning techniques implemented through Scikit-learn, including regression and clustering algorithms, are applied to identify patterns and build predictive models.

By combining exploratory data analysis with machine learning approaches, this study seeks to provide a comprehensive understanding of the factors influencing music popularity. The insights generated can assist industry stakeholders in making informed decisions regarding artist promotion, content strategy, and audience targeting. Furthermore, this research highlights the growing importance of data analytics in shaping the future of the music industry.

2 Analysis

2.1 Data Cleaning

Data cleaning and preprocessing were critical steps to ensure the integrity and reliability of the dataset before conducting any analysis. Python-based data processing techniques were employed to systematically identify and correct inconsistencies within the dataset. Initially, validation checks were performed to detect false or invalid entries, such as incorrect rank values or improperly formatted records. Special attention was given to duplicate data, where repeated song-date combinations were identified using grouping and filtering methods and subsequently removed to preserve the uniqueness of each daily playlist snapshot. Additionally, missing or inconsistent values were examined and handled appropriately to avoid bias in the analysis. These steps ensured that the dataset maintained structural consistency, eliminated redundancy, and provided a high-quality foundation for accurate feature engineering and downstream analytical tasks.

2.2 Feature Engineering

Feature engineering was performed to derive meaningful insights from the raw dataset by constructing additional analytical variables. A new metric based on the duration of songs was developed to evaluate total listening time associated with each artist, enabling a deeper understanding of artist-level engagement. Furthermore, song rankings were analyzed in relation to their duration in minutes, allowing the study of how total listening time influences chart performance. To support this analysis, visualizations were created, including a scatter plot illustrating the relationship between song rank and artist distribution, as well as pie charts representing the proportion of song types (album, single, compilation) and the contribution of top-performing artists. These engineered features and visual representations enhanced the interpretability of the dataset and facilitated more comprehensive performance analysis.

As part of the feature engineering process, several visualizations were developed to examine relationships between derived metrics and song performance. The scatter plot of popularity versus song duration reveals that the majority of high-popularity tracks are concentrated within the 2.5 to 4-minute range, indicating an optimal duration window for maximizing listener engagement. The artist-wise rank trend analysis highlights that while some artists maintain consistently high chart positions over time, others exhibit significant fluctuations, reflecting differences in stability and sustained audience interest. Furthermore, the popularity-per-minute metric provides insight into listening efficiency, showing that certain artists achieve better average ranks with relatively shorter tracks, suggesting higher engagement per unit time. Lastly, the comparison of explicit and non-explicit content based on total playtime demonstrates a near-balanced distribution, with a slight dominance of non-explicit tracks, implying broader listener accessibility and appeal.

3 Playlist Ranking Analysis

In the analysis part i calculate an new measure for various calculation and for visualization tools. This section examines the behavior and dynamics of song rankings over time within the playlist. The distribution of ranks indicates that higher-ranked positions (Top 10) tend to exhibit greater stability, while lower-ranked positions show higher turnover, suggesting frequent entry and exit of songs.

Analysis of rank movement patterns reveals two dominant trends: certain songs demonstrate rapid upward movement, often driven by viral popularity, while others show gradual changes, indicating

sustained listener engagement. Entry behavior typically involves songs debuting at mid-to-lower ranks before climbing higher, whereas exit behavior is more common among songs with high rank volatility.

Additionally, the identification of fast risers and slow decliners highlights differences in song life-cycle. Fast-rising songs tend to achieve peak positions quickly but may lack longevity, whereas slow decliners maintain their presence on the chart over extended periods despite gradual drops in rank.

3.1 Song level performance analysis

Song-level performance analysis highlights the relationship between song duration and popularity across individual tracks. The scatter plot indicates that songs within the 2.5 to 4-minute duration range consistently achieve higher popularity scores, suggesting an optimal length for audience engagement. While a few longer-duration tracks also perform well, they are relatively less frequent, and shorter or excessively long songs tend to show lower or more inconsistent popularity. This pattern suggests that songs with moderate duration are more likely to sustain listener interest, thereby contributing to better overall performance and chart success.

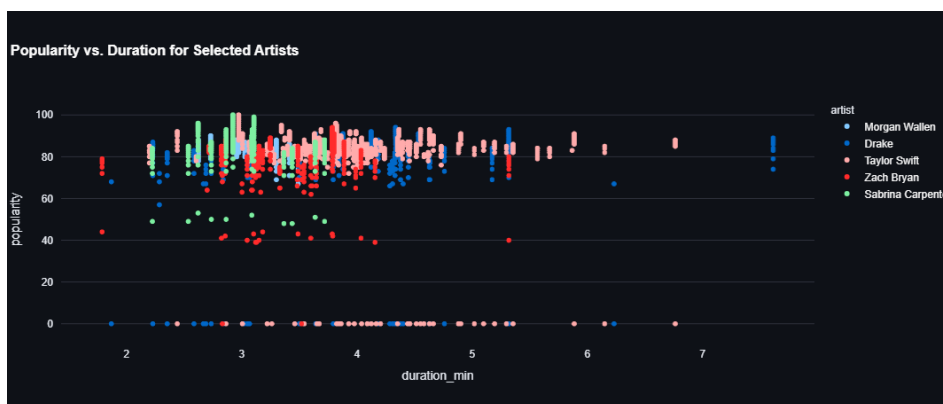


Figure 1: Song Performance

3.2 Artist Performance Analysis

Artist performance analysis based on rank trends shows clear differences in consistency and dominance among top artists. Drake consistently maintains positions within the Top 5 to Top 15 range, indicating strong and stable performance over time. Morgan Wallen and Taylor Swift also exhibit relatively stable patterns, frequently appearing within the Top 10 to Top 20 ranks, suggesting sustained listener engagement. Zach Bryan shows moderate fluctuation, with rankings typically varying between Top 10 and Top 25, reflecting periodic spikes in popularity. In contrast, some artists display higher variability, with ranks shifting more widely across the chart, indicating less consistent audience retention. Overall, artists who remain within the Top 20 with minimal fluctuation demonstrate stronger dominance and long-term success in the playlist.

3.3 Popularity score analysis

Popularity analysis based on the scatter plot reveals a noticeable clustering pattern between song duration and popularity scores, indicating a moderate positive relationship within an optimal duration window of approximately 2.5 to 4 minutes. The density of data points in this range suggests that songs of moderate length are more likely to achieve higher listener engagement. Taylor Swift distinctly stands out as the most popular artist, with multiple tracks forming a concentrated cluster in the high-popularity region, reflecting both consistency and strong audience retention. Artists such as Drake and Morgan Wallen also contribute to the upper distribution, though with comparatively higher dispersion, indicating variability in performance across their tracks. In contrast, other artists show a broader spread across mid-to-lower popularity values, highlighting weaker or less consistent engagement. Overall, the observed clustering, distribution density, and variation across artists suggest that both song duration and artist influence play a significant role in shaping popularity outcomes.

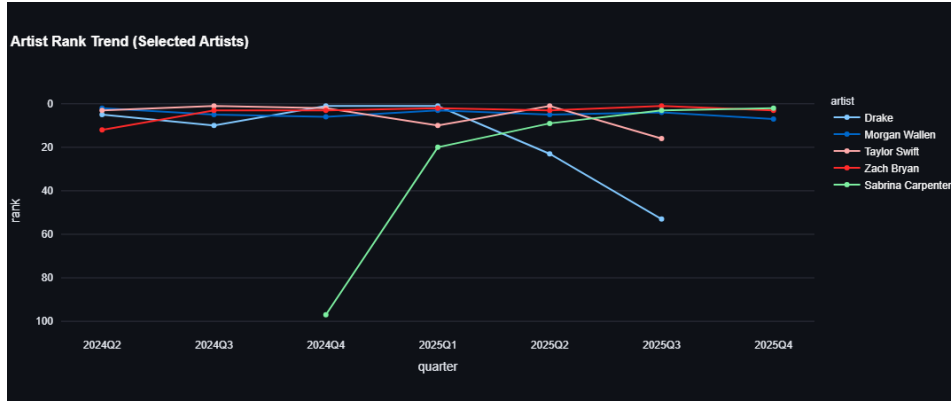


Figure 2: Top Artist by performance

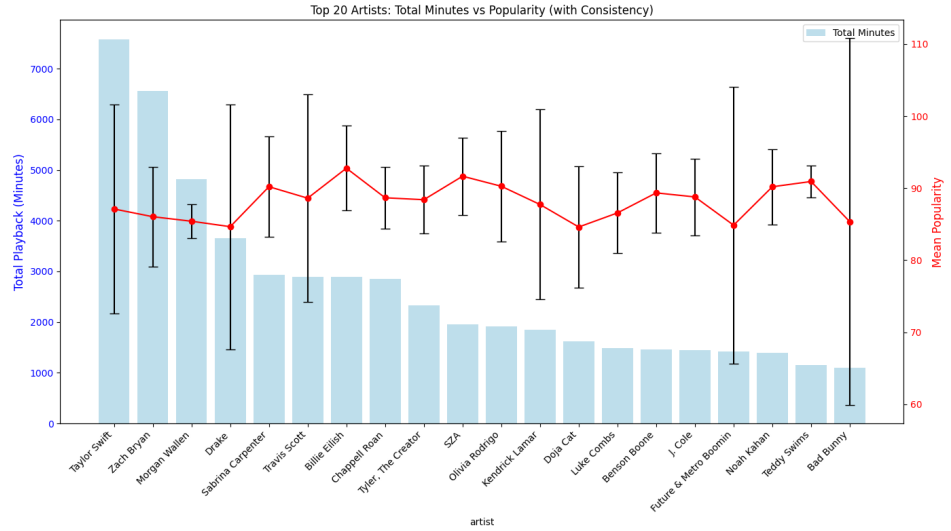


Figure 3: Popularity of Artist

3.4 Analysis by content

Content attribute analysis comparing explicit and non-explicit tracks reveals a relatively balanced distribution in total listening time, with a slight dominance of non-explicit songs. When connected with ranking and popularity trends, non-explicit tracks tend to exhibit more consistent performance, often maintaining stable positions across mid-to-upper ranks, which contributes to their higher cumulative duration. In contrast, explicit songs, while competitive in terms of popularity, show greater variability in rank positions, indicating fluctuations in listener engagement. This suggests that non-explicit content may benefit from broader audience appeal and sustained streaming behavior, whereas explicit tracks may experience more short-term spikes in popularity. Overall, the relationship between content type, rank stability, and popularity highlights that non-explicit songs are more likely to achieve consistent long-term performance within the playlist.

3.5 Top 10 artist by listening time

The table representing artist popularity by total listening time (in minutes) provides insights into cumulative audience engagement across different artists. It highlights how certain artists, particularly Taylor Swift, achieve significantly higher total listening durations, indicating both frequent playlist appearances and sustained listener interest. Artists such as Drake and Morgan Wallen also demonstrate strong performance, though with comparatively lower total minutes, reflecting either fewer entries or shorter song durations. This metric effectively combines both song duration and chart presence, offering

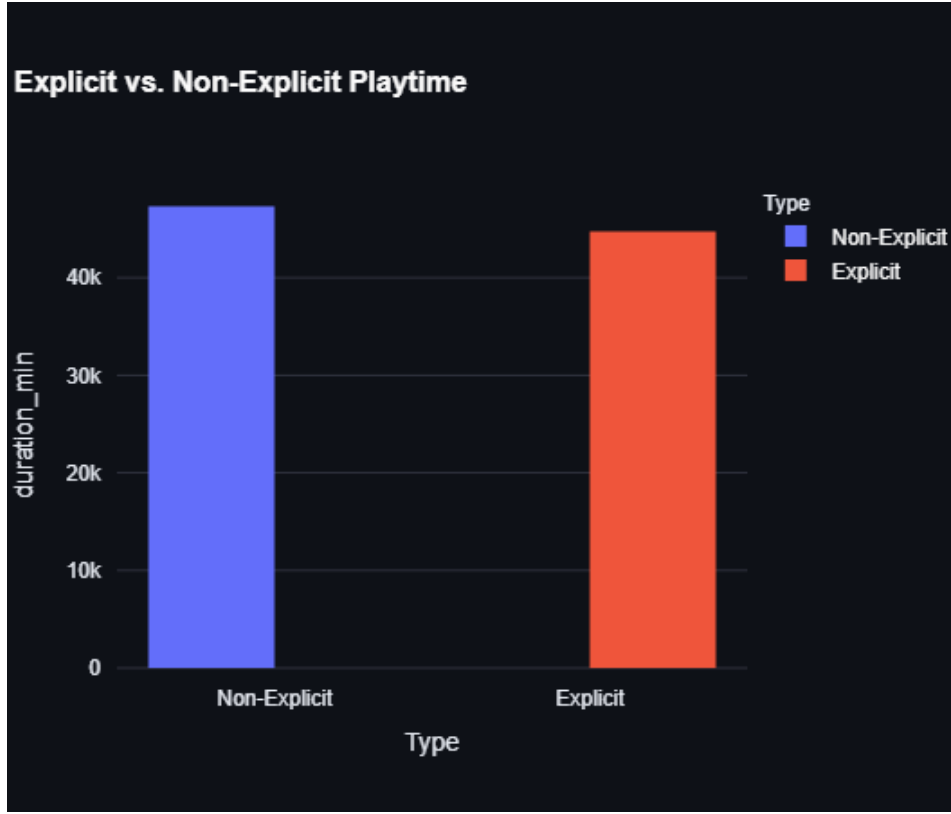


Figure 4: Explicite vs Non-explicite

a more comprehensive view of artist impact beyond individual track popularity. Overall, higher total listening time correlates with consistent chart performance and broader audience reach, reinforcing the importance of both content frequency and engagement duration in determining artist success. Table1.

3.6 Analysis by song Type

[h!] Analysis of song types based on category—album, single, and compilation—reveals distinct patterns in their contribution to overall playlist performance. Album tracks dominate the dataset, accounting for approximately 72.1% of the total, indicating that a large portion of playlist content is derived from full-length projects. Singles make up around 27.7%, and despite their smaller share, they often achieve higher visibility and stronger ranking positions due to targeted promotion strategies. Compilation tracks represent only a minimal fraction of the dataset, reflecting their limited presence and impact. Overall, while album tracks contribute the majority of content, singles tend to play a more significant role in driving popularity and chart success.Figure5.

Sr.no.	Artist	Popularity	Position	Song
1	Dominic Fike	68.568	47.5	2
2	Rithy Mitch	61.3546	43.19	56
3	The Kid LAROI	60.028	31.68	149
4	Lil Mabu	58.774	27.92	14
5	Nicki Minaj and ice spice	50.30	18.5	104
6	Chino Pascas	49.080	43.06	32
7	Nat King cole	48.154	27.92	65
8	Brent Faiyaz	46.612	43.38	18
9	Paul Russell	45.696	36	1
10	Dean martin	45.526	17.11	45

Table 1: Popularity by Time(Minutes).

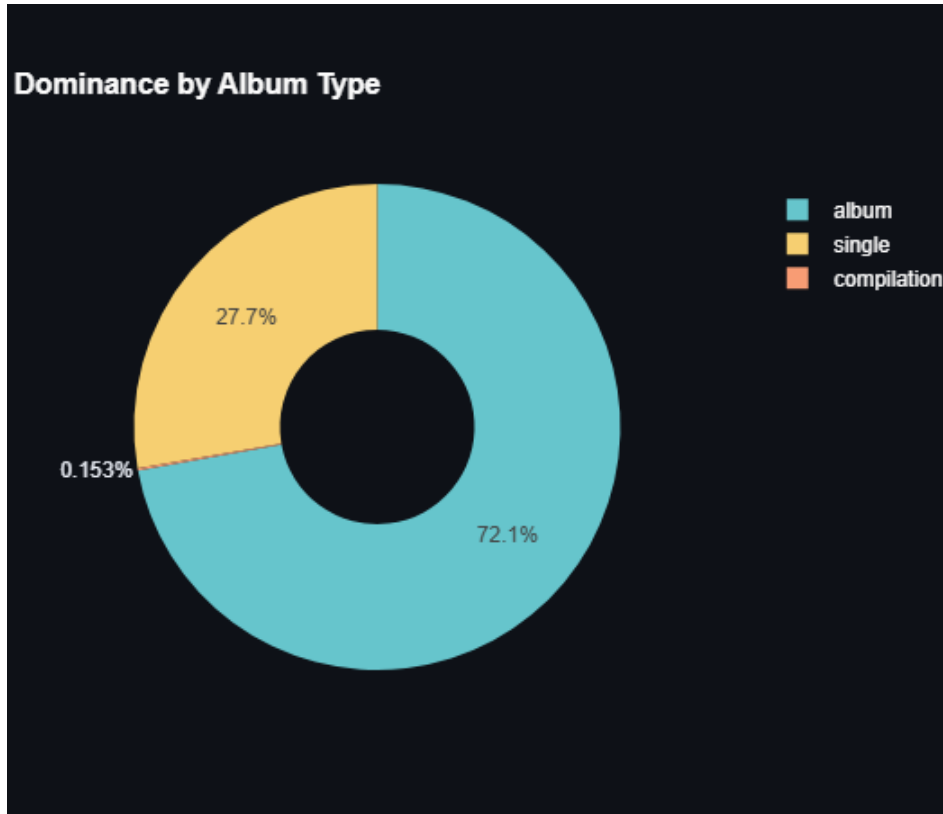


Figure 5: Dominance of artist by type

4 Conclusion

In conclusion, this study provides a comprehensive analysis of song performance, artist influence, and content characteristics within the Atlantic Music US playlist. By leveraging data cleaning, feature engineering, and visual analytics, key patterns in popularity and ranking behavior were identified. The findings indicate that songs with moderate duration tend to achieve higher popularity and stability, while singles, despite their lower proportion, often outperform album and compilation tracks in terms of ranking success. Artist-level analysis highlights the dominance of performers such as Taylor Swift, Drake, and Morgan Wallen, who consistently maintain strong chart positions and high cumulative listening time. Additionally, non-explicit content demonstrates slightly more stable performance, suggesting broader audience appeal. Overall, the study emphasizes that both structural attributes of songs and artist influence play a critical role in determining chart success. Future research can extend this analysis by incorporating predictive modeling, genre-based comparisons, and cross-platform data to gain deeper insights into music consumption patterns.

References

- 1] Data of Atlantic Music USA.